

FIT1058 — Chapter 2: Functions

Exercise Solutions

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Exercise 1

Let A be a finite set. How many indicator functions with domain A are there?

Solution. An indicator function is defined by

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Let $|A| = n$. The number of functions $\chi : A \rightarrow \{0, 1\}$ equals the number of ways to independently assign each of the n elements of A a value in $\{0, 1\}$, which is 2^n .

Equivalently, each such function corresponds to exactly one subset $X \subseteq A$ (namely $X = \{x \in A : \chi(x) = 1\}$), so the number of indicator functions equals $|\mathcal{P}(A)| = 2^n$. ■

Exercise 2

Let A and B be any sets. Express the indicator functions of each of the following in terms of the indicator functions χ_A and χ_B .

- (a) $\overline{A}$
- (b) $A \cap B$
- (c) $A \setminus B$
- (d) $A \triangle B$

Solution. (a)

$$\chi_{\overline{A}}(x) = 1 - \chi_A(x).$$

(b) $x \in A \cap B \iff x \in A \wedge x \in B$, so

$$\chi_{A \cap B} = \chi_A \cdot \chi_B.$$

(c) $x \in A \setminus B \iff x \in A \wedge x \notin B$, so

$$\chi_{A \setminus B} = \chi_A \cdot \chi_{\overline{B}} = \chi_A \cdot (1 - \chi_B) = \chi_A - \chi_A \chi_B.$$

(d) Since $A \triangle B = (A \setminus B) \cup (B \setminus A)$ is a disjoint union,

$$\chi_{A \triangle B} = \chi_{A \setminus B} + \chi_{B \setminus A} = (\chi_A - \chi_A \chi_B) + (\chi_B - \chi_A \chi_B) = \chi_A + \chi_B - 2\chi_A \chi_B.$$

■

Exercise 3

If two functions $f : A \rightarrow \mathbb{R}$ and $g : A \rightarrow \mathbb{R}$ have the same domain A and give real numbers as their values, then their sum $f + g : A \rightarrow \mathbb{R}$ is defined for all $x \in A$ by

$$(f + g)(x) = f(x) + g(x).$$

What is the sum of all the indicator functions of all the subsets of a finite set?

Proof. Let A be a finite set with $|A| = n$. Since function addition is defined pointwise, for any $x_i \in A$,

$$\left(\sum_{X \subseteq A} \chi_X \right) (x_i) = \sum_{X \subseteq A} \chi_X(x_i).$$

Since $\chi_X(x_i) = 1$ if $x_i \in X$ and 0 otherwise, this sum counts the number of subsets $X \subseteq A$ containing x_i .

Method 1 (direct count). A subset containing x_i is formed by fixing x_i to be included, and choosing freely, independently, whether each of the remaining $n - 1$ elements is included. This gives 2^{n-1} such subsets.

Method 2 (summing over subset sizes). For each size $k = 1, \dots, n$, the number of size- k subsets containing x_i is obtained by fixing x_i as one member and choosing the remaining $k - 1$ members from the other $n - 1$ elements: $\binom{n-1}{k-1}$. Summing over all sizes,

$$\sum_{X \subseteq A} \chi_X(x_i) = \sum_{k=1}^n \binom{n-1}{k-1} = \sum_{j=0}^{n-1} \binom{n-1}{j} = 2^{n-1},$$

using the binomial identity $\sum_{j=0}^m \binom{m}{j} = 2^m$ with $m = n - 1$.

Both methods agree: $\left(\sum_{X \subseteq A} \chi_X \right) (x_i) = 2^{n-1}$ for every $x_i \in A$, independent of which x_i is chosen. So $\sum_{X \subseteq A} \chi_X$ is the constant function on A with value 2^{n-1} . ■

Exercise 5

Functions can be viewed as sets of ordered pairs, so we can combine functions using set operations. The result will be a binary relation but, depending on the operation, it may or may not be another function.

Suppose $f : A \rightarrow B$ and $g : C \rightarrow D$ are functions. Which of the following is always a function?

$$f \cap g; \quad f \cup g; \quad f \setminus g; \quad g \setminus f; \quad f \triangle g; \quad f \times g.$$

For each that is always a function, give its definition in the usual form, including showing how the rule depends on f and g . For each that is not necessarily a function, explain why this is the case (e.g., with the help of examples for f and g under which the operation does not give a function).

Proof. Recall that $R \subseteq A \times B$ is a function if and only if R satisfies both:

- **Existence:** $\{x \in A : (x, y) \in R \text{ for some } y \in B\} = A$;
- **Uniqueness (well-definedness):** for every $x \in A$, the set $R(x) = \{y \in B : xRy\}$ contains exactly one element.

(i) $f \cap g$.

$$(f \cap g)(x) = \{f(x)\} \cap \{g(x)\}.$$

When $f(x) = g(x)$: $(f \cap g)(x) = \{f(x)\}$. When $f(x) \neq g(x)$: $(f \cap g)(x) = \emptyset$ — does not exist. So $f \cap g$ is *not necessarily* a function; it can fail **existence**.

(ii) $f \cup g$.

$$(f \cup g)(x) = \{f(x)\} \cup \{g(x)\}.$$

When $f(x) = g(x)$: $|(f \cup g)(x)| = 1$. When $f(x) \neq g(x)$: $|(f \cup g)(x)| = 2$. So $f \cup g$ is *not necessarily* a function; it can fail **uniqueness**.

(iii) $f \setminus g$.

$$(f \setminus g)(x) = \{f(x)\} \setminus \{g(x)\}.$$

When $f(x) = g(x)$: $|f \setminus g| = 0$ — does not exist. When $f(x) \neq g(x)$: $|f \setminus g| = 1$. So $f \setminus g$ is *not necessarily* a function; it can fail **existence**.

(iv) $g \setminus f$. By the same argument as (iii) with the roles of f, g interchanged, $g \setminus f$ is *not necessarily* a function.

(v) $f \triangle g$.

$$(f \triangle g)(x) = \{(f \setminus g)(x)\} \cup \{(g \setminus f)(x)\}.$$

When $f(x) = g(x)$: $|f \triangle g| = 0$ — does not exist. When $f(x) \neq g(x)$: $|f \triangle g| = 2$. So $f \triangle g$ is *not necessarily* a function; depending on f, g , it can fail either **existence** or **uniqueness**.

(vi) $f \times g$.

$$f \times g = \{(a, b), (c, d) : (a, b) \in f, (c, d) \in g\}.$$

This defines a function

$$f \times g : A \times C \rightarrow B \times D, \quad (f \times g)(x, y) = (f(x), g(y)).$$

Existence: for all $(x, y) \in A \times C$, since f and g are functions, $f(x)$ and $g(y)$ must exist, so $(f(x), g(y))$ exists. **Uniqueness:** since $f(x)$ and $g(y)$ are each unique, the pair $(f(x), g(y))$ is unique. So $f \times g$ is *always* a function. ■

Exercise 6

Suppose that $f : A \rightarrow B$ and $g : C \rightarrow D$ are functions with disjoint domains: $A \cap C = \emptyset$. Is the disjoint union $f \sqcup g$ always a function? If so, give its definition in terms of f and g . If not, why not?

Solution. By 5(ii) $f \cup g$ is always a function. ■

Exercise 8

A bitstring is a string over the binary alphabet $\{0, 1\}$; in other words, it is a member of $\{0, 1\}^*$.

(a) Give a bijection from $\{0, 1\}^*$ to $\mathbb{N}$.

Now let $\{0, 1\}^{*+}$ denote the set of all finite nonempty tuples (or sequences) of bitstrings. Examples of members of $\{0, 1\}^{*+}$ include:

$$(011, 1, 10001), \quad (10, 10, 10, 10), \quad (\varepsilon, 1011), \quad (1010010001).$$

(b) (Challenge) Give a bijection from $\{0, 1\}^{*+}$ to $\{0, 1\}^*$.

Solution. (a)

Setup: grouping by length. For each $n \geq 0$, there are 2^n bitstrings of length n . We assign each bitstring a natural number (“index”) as follows:

length	bitstring	number of bitstrings	index $\in \mathbb{N}$
0	ε	1	0
1	0, 1	$2 = 2^1$	1, 2
2	00, 01, 10, 11	2^2	3, 4, 5, 6
$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	$\dots$	2^n	$\dots$

Defining f_n . For $\{0, 1\}^n$, the value of a string is a member of $\{0, 1, \dots, 2^n - 1\}$. Define f_n to be the value of $\{0, 1\}^n$:

$$f_n : \{0, 1\}^n \longrightarrow \{0, 1, \dots, 2^n - 1\}, \quad f_n(s) = f_n(s_1 s_2 \dots s_n) = \sum_{i=1}^n s_i \cdot 2^{n-i}.$$

Claim 0.1. For every n , f_n is a bijection.

Proof. When $n = 0$: $f_0 : \{\varepsilon\} \rightarrow \{0\}$, $f_0(\varepsilon) = 0$, is a bijection.

Assume $f_{n-1} : \{0, 1\}^{n-1} \rightarrow \{0, \dots, 2^{n-1} - 1\}$ is a bijection.

For f_n ,

$$f_n(s_1 \dots s_n) = s_1 \cdot 2^{n-1} + \sum_{i=2}^n s_i \cdot 2^{n-i} = s_1 \cdot 2^{n-1} + f_{n-1}(s_2 \dots s_n).$$

When $s_1 = 0$: $f_n(s_1 \dots s_n) \in \{0, \dots, 2^{n-1} - 1\}$.

When $s_1 = 1$: $f_n(s_1 \dots s_n) \in \{2^{n-1}, \dots, 2^{n-1} - 1 + 2^{n-1}\} = \{2^{n-1}, \dots, 2^n - 1\}$.

We know that $f_n(s_1 \dots s_n) \in \{0, \dots, 2^n - 1\}$.

By induction, we know that f_n is a bijection. ■

Starting index of each length group. Observe that, for length = n , the starting index is equal to the number of bitstrings of length = 0 to length = $n - 1$:

$$\text{starting index} = \sum_{k=0}^{n-1} |\{0, 1\}^k| = \sum_{k=0}^{n-1} 2^k = \frac{2^n - 1}{2 - 1} = 2^n - 1.$$

So, $f : \{0, 1\}^* \rightarrow \mathbb{N}$ is equal to $(2^n - 1) +$ (position of bitstring in the group of the length):

$$f(s) = (2^n - 1) + f_n(s), \quad n = |s|.$$

Claim 0.2. f is a bijection.

Proof. We know $f(s) \in [2^{|s|} - 1, 2^{|s|+1} - 2]$.

First, we prove f is injective.

Case 1: $s \neq t$ and $|s| \neq |t|$. Let $|s| < |t|$. Then

$$f(s) \in [2^{|s|} - 1, 2^{|s|+1} - 2], \quad f(t) \in [2^{|t|} - 1, 2^{|t|+1} - 2].$$

Since $|s| < |t|$ are integers, $|s| + 1 \leq |t|$, so

$$2^{|s|+1} - 2 \leq 2^{|t|} - 2 < 2^{|t|} - 1.$$

So $[2^{|s|} - 1, 2^{|s|+1} - 2] \cap [2^{|t|} - 1, 2^{|t|+1} - 2] = \emptyset$. Hence $f(t) \neq f(s)$.

Case 2: $s \neq t$ and $|s| = |t|$.

$$f_{|s|}(s) \neq f_{|t|}(t) \implies f(s) \neq f(t).$$

By Case 1 and 2, f is injective.

Now we prove f is surjective, i.e. need to prove $\forall m \in \mathbb{N}, \exists s \in \{0, 1\}^*, f(s) = m$.

We know $f(s) \in [2^{|s|} - 1, 2^{|s|+1} - 2]$, so we need

$$2^{|s|} - 1 \leq m \leq 2^{|s|+1} - 2 \iff 2^{|s|} \leq m+1 \leq 2^{|s|+1} - 1 \iff 2^{|s|} \leq m+1 < 2^{|s|+1} \iff |s| \leq \log_2(m+1) < |s|+1.$$

So let $n = \lfloor \log_2(m+1) \rfloor$. Then

$$2^n \leq m+1 < 2^{n+1} \implies 0 \leq m - 2^n + 1 < 2^n.$$

So $m - 2^n + 1 \in [0, 2^n) \subseteq [0, 2^n - 1]$, which is the domain of f_n^{-1} (i.e., $m - 2^n + 1$ is in the range of f_n).

f_n is surjective, so $\exists s \in \{0, 1\}^n$ such that $f_n(s) = m - 2^n + 1$. Then

$$f(s) = (2^n - 1) + f_n(s) = (2^n - 1) + (m - 2^n + 1) = m.$$

We proved that $\forall m \in \mathbb{N}, \exists s \in \{0, 1\}^*, f(s) = m$. So, f is surjective.

Hence, f is a bijection. ■

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Exercise 9

Let A be a set of n elements and let $x \in A$. We say that a function $f : A \rightarrow A$ *fixes* x if $f(x) = x$. We also say in this case that x is a *fixed point* of f .

This says nothing about what f does to other elements of A ; it is possible that some other elements are fixed by f too, or that none are. All this definition requires is that f sends x to itself.

Now let $X \subseteq A$. The function f *fixes* X if it fixes all elements of X . So, for all $x \in X$, we have $f(x) = x$. In other words, the restriction of f to X is just the identity function on X :

$$f|_X = i_X.$$

Members of $A \setminus X$ may or may not be fixed by f .

Let $\text{Fix}(X)$ be the set of bijections on A that fix X , and let F be the set of all bijections on A that fix at least one element of A .

Using Exercise 1.13 (i.e., Exercise 13 in Chapter 1), express $|F|$ in terms of the set sizes $|\text{Fix}(X)|$ using all $X \subseteq A$.

Solution. ■

Exercise 10

For each $k \in \{1, \dots, n\}$ and any subset $X \subseteq A$ with $|X| = k$, write f_k for the number of bijections on A that fix X .

(a) Why don't we include X in the notation f_k , given that its definition refers to X ?

(b) Give an expression for f_k .

- (c) Express $|F|$ in terms of $f_1, f_2, \dots, f_n$, and then use your expression for f_k to give an expression for $|F|$.
- (d) Hence express the number of fixed-point-free bijections on A in terms of $f_1, f_2, \dots, f_n$. A function is fixed-point-free if it has no fixed points.
- (e) For each of the following sets, what proportion of all bijections on the set are fixed-point-free? $\{0, 1\}$; $\{\spadesuit, \clubsuit, \diamondsuit, \heartsuit\}$; the set of ten decimal digits; the 26-letter English alphabet.
- (f) What is the largest set for which you can determine this proportion, and what is the value of the proportion for that set?

Solution. (a)

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Solution. (b)

■

Solution. (c)

■

Solution. (d)

■

Solution. (e)

■

Solution. (f)

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Exercise 11

- (a) Write down all surjections from $\{1, 2, 3, 4\}$ to $\{a, b, c\}$. Compare your answer with Exercise 1.20.
- (b) Define a surjection from
- $$\{\text{surjections from } \{1, 2, 3, 4\} \text{ to } \{a, b, c\}\}$$
- to
- $$\{\text{partitions of } \{1, 2, 3, 4\} \text{ into three parts}\}.$$
- (c) What sizes can the preimages of members of its image be?

Solution.

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Exercise 12

Suppose $f : A \rightarrow B$ and $g : C \rightarrow D$ are bijections, $A \cap C = \emptyset$ and $B \cap D = \emptyset$. Since $A \cap C = \emptyset$, the disjoint union $f \sqcup g$ exists. Show that $f \sqcup g$ is a bijection and express the inverse of $f \sqcup g$ in terms of the inverses of f and g .

Solution.

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Exercise 13

If the function f has an inverse function, which of its iterated compositions $f^{(k)}$ have inverse functions (where $k \in \mathbb{N}$)?

Solution.

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Exercise 14

Determine all functions from $\{1, 2, 3\}$ to itself whose indefinitely iterated composition is a constant function (i.e., $f^{(k)}$ is a constant function if k is large enough).

For a challenge: investigate when this happens in general. Try to characterise those functions $f : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ such that, for large enough k , the iterated composition $f^{(k)}$ is constant.

Solution.

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Exercise 15

This exercise is about the Caesar slide cryptosystem, one of the oldest and simplest cryptosystems. It is not secure, but its core operation is used in many stronger and more complex cryptosystems.

Caesar slide encrypts a message by sliding each letter along the alphabet, rightwards, by some fixed number of steps. This fixed number is the key, k , and the same amount of sliding is done to each letter of the message, with wrap-around.

Formal definition: the message space M is the set of all strings of English lower case letters. The key space K is the English lower-case alphabet $\{a, \dots, z\}$. The cypher space C equals M . Letter addition $\alpha + \beta$ slides α rightwards by β steps, with wrap-around (and letter subtraction slides leftwards).

The encryption function $e : M \times K \rightarrow M$ is defined for $m = m_1 m_2 \dots m_n \in M$ and $k \in K$ by

$$e(m, k) = c_1 c_2 \dots c_n, \quad c_i = m_i + k.$$

The decryption function $d : M \times K \rightarrow M$ is defined for $c = c_1 c_2 \dots c_n \in M$ and $k \in K$ by

$$d(c, k) = m_1 m_2 \dots m_n, \quad m_i = c_i - k.$$

Referring to the definitions of *embeddable*, *equivalent* and *idempotent* given in Assignment 1: prove that the Caesar slide cryptosystem is idempotent.

Solution.

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Exercise 16

If f is an injection and f^{-1} is its inverse relation, what can you say about $f \circ f^{-1}$ and $f^{-1} \circ f$? What kinds of functions or relations are they, and what are their domains and codomains?

Solution.

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Exercise 18

Recall the Parent relation from §2.13. Suggest a good name for its transitive closure. Then use that relation (i.e., the transitive closure) to define formally the relation that links two people that are related to each other (however distantly).

Solution.

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Exercise 19

Rephrase the Collatz Conjecture as a statement about the transitive closure of the Collatz function.

Solution.

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Exercise 20

Below are some binary relations on the set of all Python programs, denoted P, Q , with relations $\simeq_1, \dots, \simeq_{10}$. For each relation, determine whether or not it is an equivalence relation, and give reasons.

- | | |
|-------------------|---|
| $P \simeq_1 Q$ | P and Q have the same number of characters |
| $P \simeq_2 Q$ | P and Q compute the same function |
| $P \simeq_3 Q$ | P and Q have the same set of variable names |
| $P \simeq_4 Q$ | P and Q have at least one variable name in common |
| $P \simeq_5 Q$ | P and Q have no variable names in common |
| $P \simeq_6 Q$ | P and Q were written by the same set of programmers |
| $P \simeq_7 Q$ | P and Q have at least one coauthor in common |
| $P \simeq_8 Q$ | P was completed before Q |
| $P \simeq_9 Q$ | P was completed on the same day as Q |
| $P \simeq_{10} Q$ | P was completed after Q |

Solution.

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Exercise 22

The anagram relation on the set of all English words is defined as follows. Let $x_1x_2 \cdots x_m$ and $y_1y_2 \cdots y_n$ be two English words of lengths m and n respectively. The ordered pair $(x_1 \cdots x_m, y_1 \cdots y_n)$ belongs to the anagram relation if and only if there exists a bijection $f : \{1, \dots, m\} \rightarrow \{1, \dots, n\}$ such that, for all $i \in \{1, \dots, m\}$, $x_i = y_{f(i)}$.

- Find the largest set of three-letter words you can that are all related to each other by the anagram relation.
- For each $k = 1, 2, 3, 4$, find an English word of length k that is not related to any other word by anagram.
- Prove that anagram is an equivalence relation.

Solution. (a)

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Solution. (b)

■

Solution. (c)

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Exercise 24

Recall that when you have a table with n columns, its rows represent the n -tuples in an n -ary relation. For $1 \leq i \leq n$, let A_i be a set containing all the entries in column i (and possibly more elements), so that the columns of the table define an n -ary relation on $A_1 \times A_2 \times \cdots \times A_n$. Assuming the columns of this table are all distinct, how many ternary relations can you construct by choosing columns from this table of n columns?

Solution.

